MME304

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY

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Level-Term: 3-2

Course Title: Metal Joining and Forming Processes Laboratory

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Abstract

This experiment focused on the analysis of MIG welding and its effects on the workpiece. I conducted a series of tests, beginning with a software-based analysis and followed by the examination of the macrostructure and microstructures of each sample part. A hardness profile was then constructed for each specimen, with a comparison drawn based on the observed microstructure. Additionally, a detailed microstructural analysis was carried out. Using the OES data, I calculated the Carbon Equivalent (CE), UNS number, and ASTM standard for the specimen. In the mechanical property testing phase, I performed tensile, nick bend, root bend, and face bend tests, organizing and analyzing the data. The overall objective of this experiment was to explore how MIG welding affects the properties of the workpiece, and to better understand the impact of these processes through mechanical testing and microstructural analysis.

Introduction:

This lab report investigates the effects of Gas Metal Arc Welding (GMAW), commonly referred to as MIG welding, on mild steel. The primary focus is on understanding the mechanical and microstructural changes induced during the welding process. This study explores the macrostructure, microstructure, and hardness profiles of the welded sample through comprehensive analysis techniques, including Rockwell Hardness testing and Optical Emission Spectrometry (OES). The findings aim to correlate these properties with specific welding parameters, offering insights into how welding conditions influence material performance in industrial applications.

Equipment List:

1. Mig welded sample
2. Hacksaw and electric saw
3. Slide calipers
4. OES machine
5. Emery paper
6. Wet polishing machine
7. Alumina powder
8. 2% and 10% nital solution
9. Optical microscope
10. HRB hardness testing machine
11. ImageJ software
12. Photoshop software
13. SolidWorks software

Experimental Procedure:

Welding the Sample (GMAW Process):

- Cleaned the metal surface with a wire brush to remove dirt, oil, and contaminants.
- Set up the welding machine with power settings of 20 V and 100A DCEP.
- Secured the welding wire into the torch.
- Turned on the shielding gas and adjusted the flow rate.
- Held the welding torch at a consistent angle and applied a straight bead onto a mild steel plate.
- Controlled the wire feed and adjusted power settings using the trigger.
- Released the trigger and allowed the workpiece to cool after welding.

Preparing the Sample for Microstructural Analysis & OES:

- Cut a 9 mm section of the sample using an hack-saw for microstructural analysis.

- Reserved a 30 mm section for Optical Emission Spectrometry (OES).
- For OES:
  -Both sides are flattened carefully using grinding machine
  -To be more fit to the OES machine, at the very edges are cut off.
- Polished the OES sample with 120 emery paper to give it a more flat shape.
- For microstructural analysis:
  -Wet-polished the sample with emery paper (120, 320, 600, 800,1200, 1500).
  -Shifted the polishing direction by 90° after each grit.
  -Used aluminum oxide powder for wet polishing.
- Etched the sample with 2% nital solution to reveal grain boundaries, for microstructure and 10% nitral for macrostructure.
- Captured microstructure images at 500x magnification.

Measuring Hardness:

- Marked specific zones on the sample for the Rockwell hardness test (HRB scale).
- Applied a minor load followed by a major load using the indenter. (By the advice of lab instructor)
- Maintained the load for a dwell time to ensure full penetration.
- Calculated the hardness value based on the depth of the indentation.
- Recorded the hardness value as HRB on the device.

Result and Calculations:

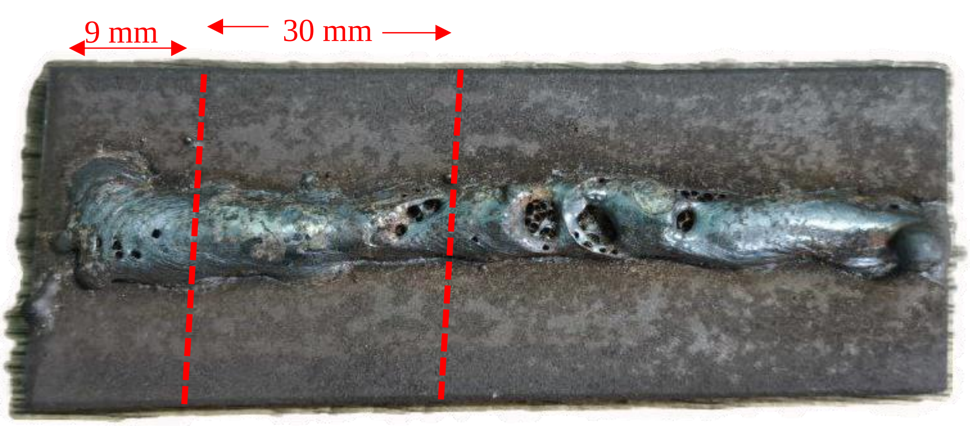


Figure 1: My MIG welding sample

SolidWorks 3D and 2D render:

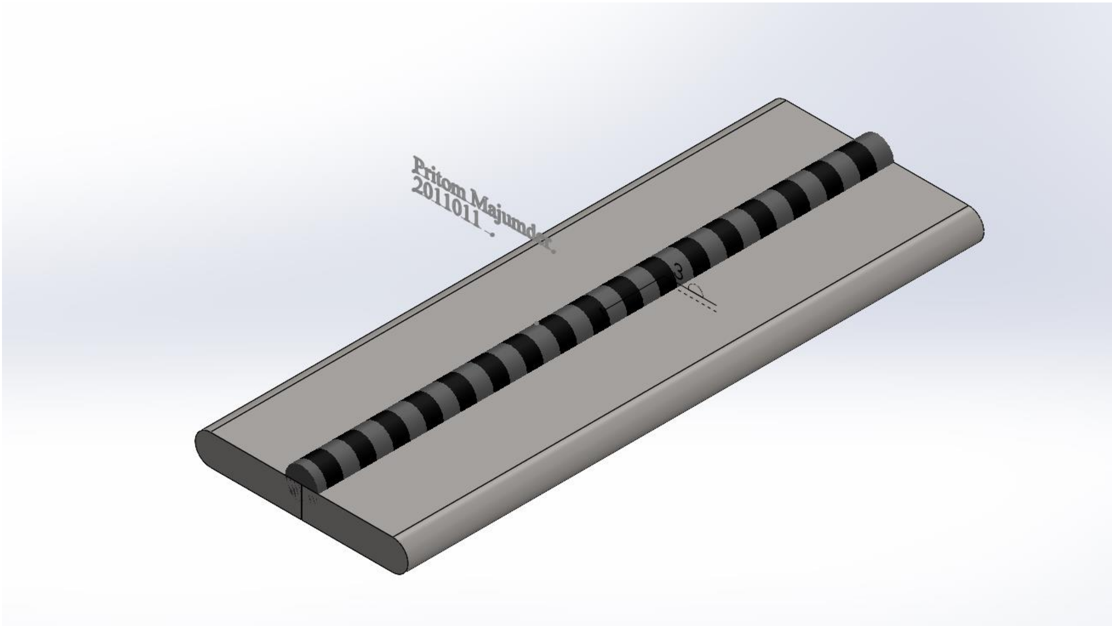


Figure 2: Isometric view

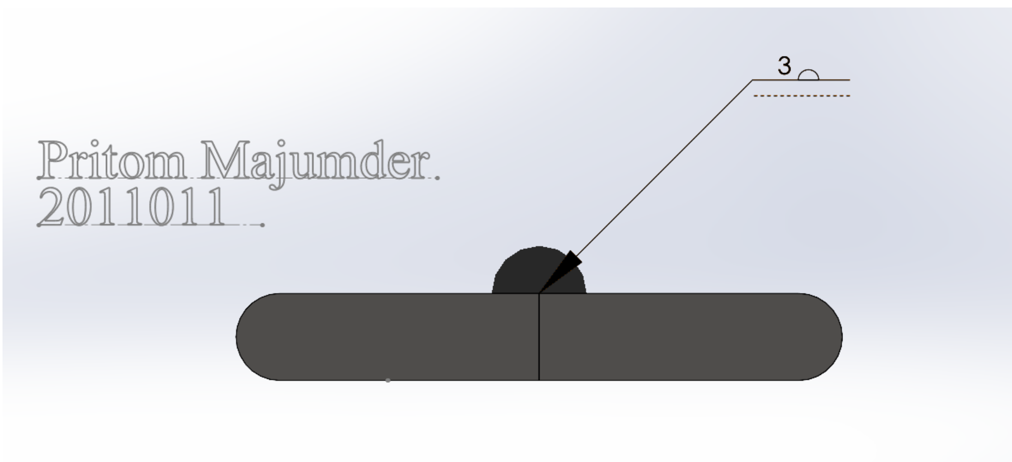


Figure 3: Side view

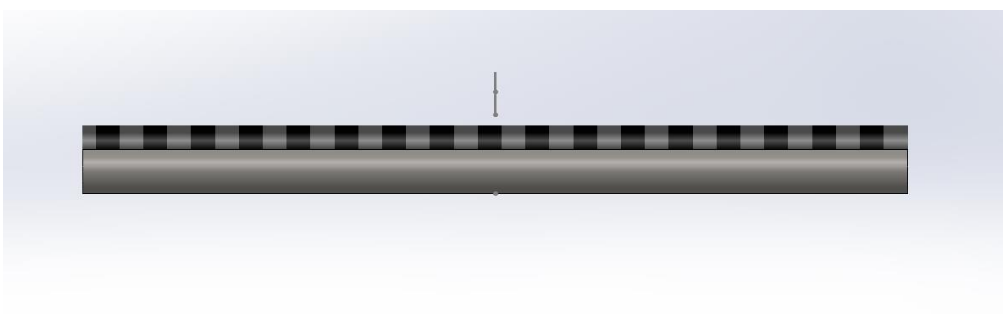


Figure 4: Right hand side view

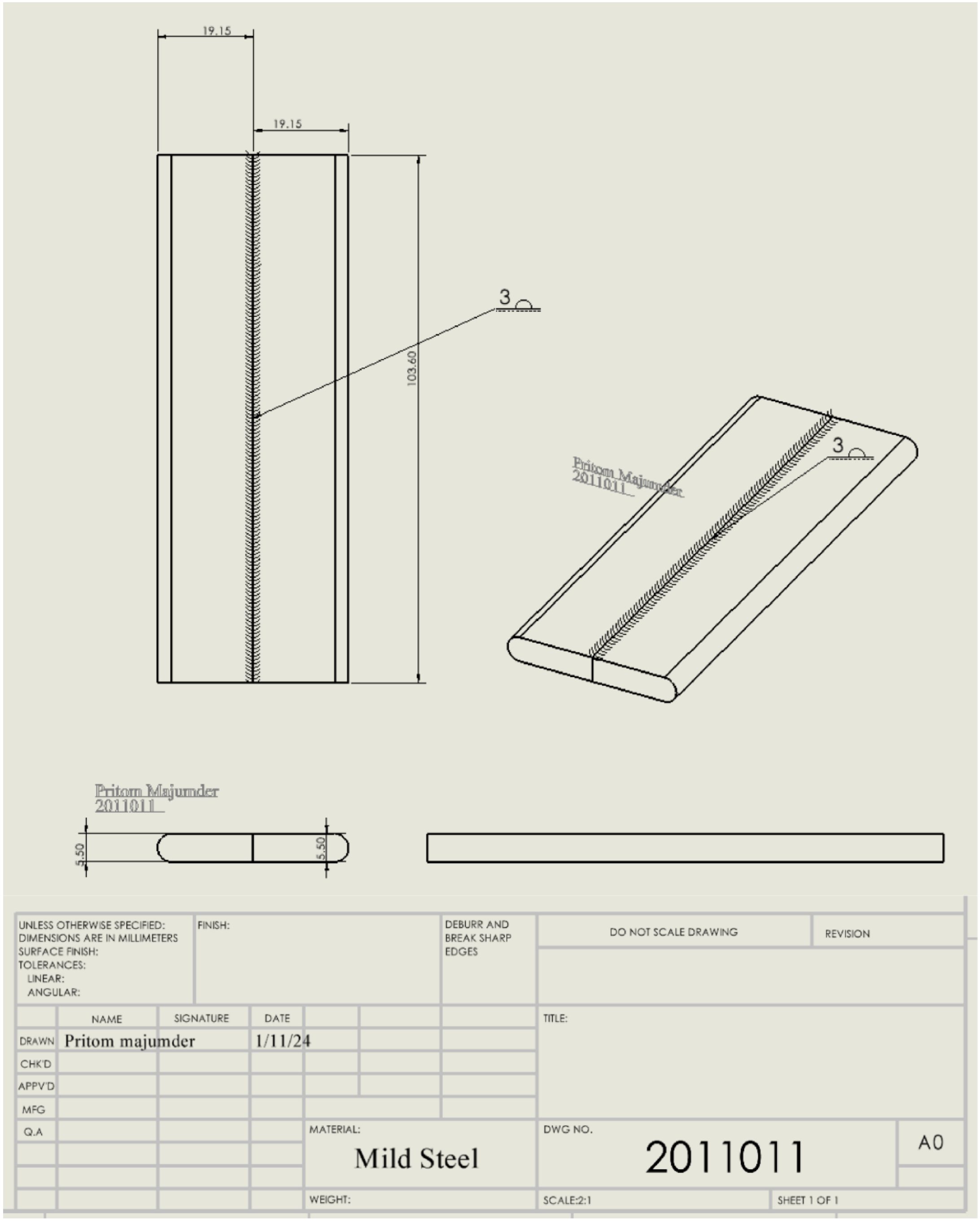
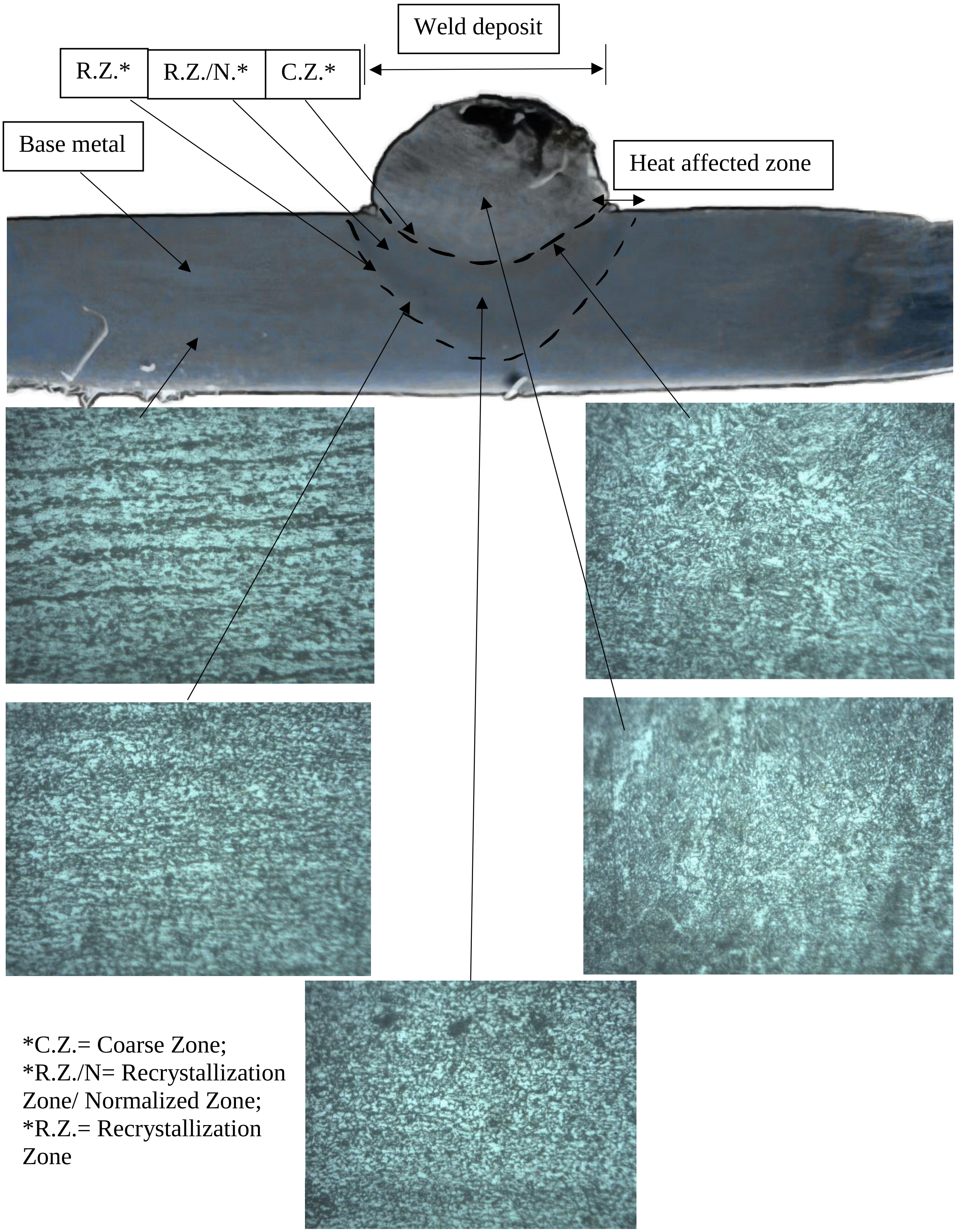


Figure 5: SolidWorks 2D render

SolidWorks Files: (I had used SolidWorks 2024 to render my model)

[important files]

Microstructure and Macrostructure:



*C.Z.= Coarse Zone;
*R.Z./N= Recrystallization Zone/ Normalized Zone;
*R.Z.= Recrystallization Zone

Analysis of microstructure:

Base metal:

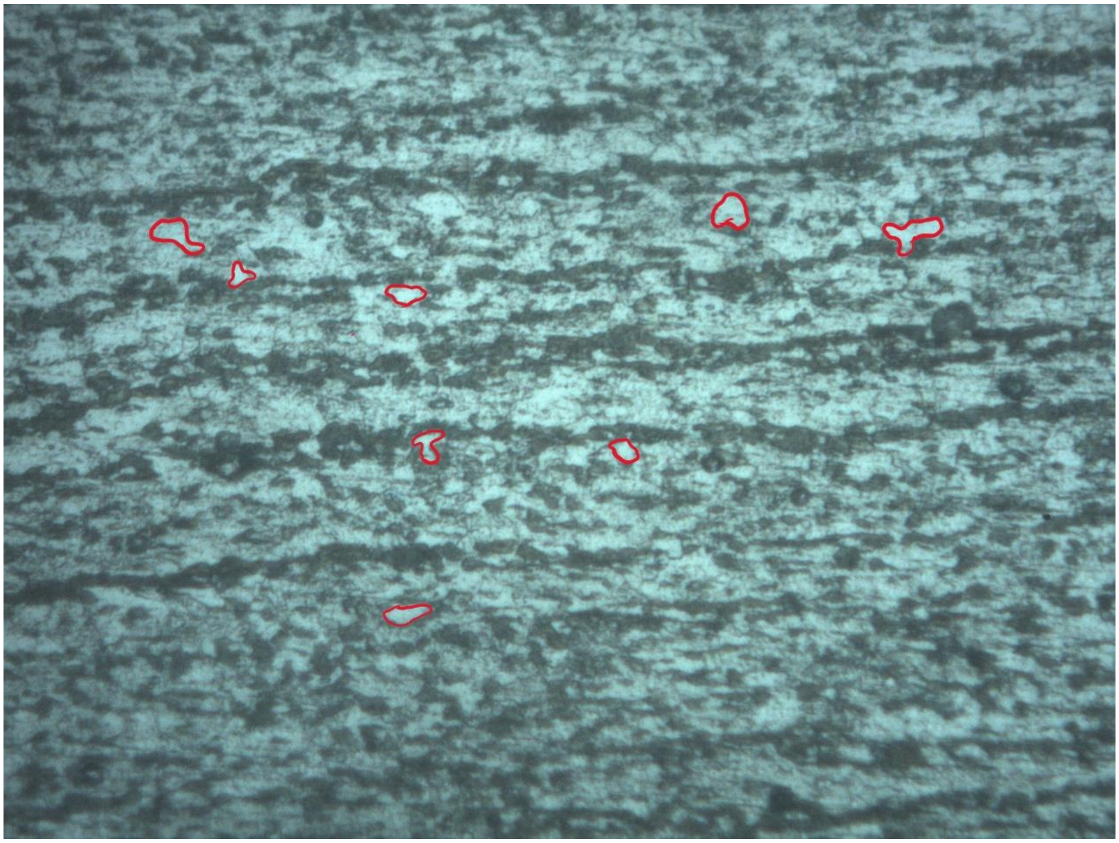


Figure 6: Area of 8 grains are calculated

	Area(um^2)	Mean	StdDev	Min	Max	Circ.	AR	Round	Solidity
1	0.134	0	25.2	0	0	0.684	1.396	0.716	0.91
2	0.408	150.851	22.692	88	198	0.362	2.429	0.412	0.719
3	0.172	166.449	31.748	70	219	0.435	1.236	0.809	0.766
4	0.608	184.7	30.751	77	229	0.472	1.067	0.937	0.759
5	0.203	166.395	26.089	100	214	0.57	1.786	0.56	0.868
6	0.208	181.987	28.602	85	229	0.419	1.344	0.744	0.709
7	0.246	151.492	23.993	72	191	0.441	2.43	0.412	0.881
8	0.462	143.317	31.696	47	195	0.641	1.387	0.721	0.899
	0.305125								

Average grain size: 0.305125 um^2

Recrystallization Zone:

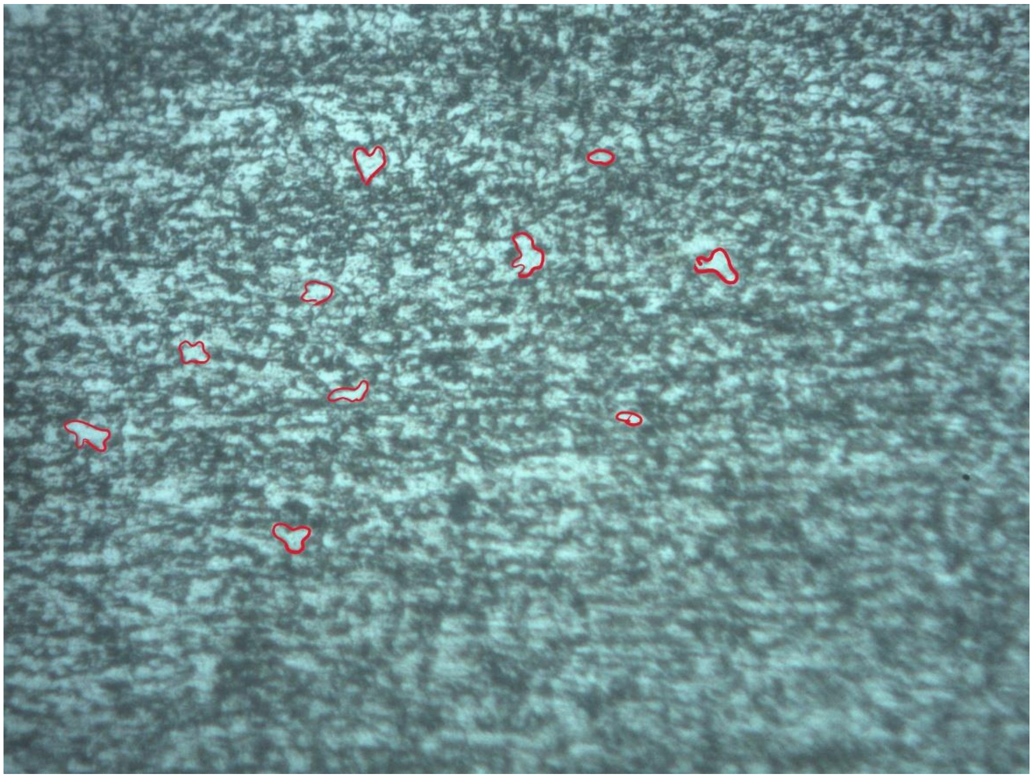


Figure 7: Area of 10 grains are calculated

	Area(um^2)	Mean	StdDev	Min	Max	Circ.	AR	Round	Solidity
1	0.299	198.892	33.379	102	240	0.574	1.796	0.557	0.869
2	0.165	195.489	24.288	112	232	0.434	1.823	0.549	0.75
3	0.145	193.909	29.876	104	237	0.479	2.134	0.469	0.809
4	0.414	194.865	26.728	119	234	0.428	1.688	0.593	0.721
5	0.356	202.114	27.328	113	240	0.424	1.102	0.908	0.725
6	0.443	190.643	30.432	111	234	0.36	2.096	0.477	0.717
7	0.427	171.03	30.599	90	219	0.424	1.836	0.545	0.714
8	0.595	175.334	28.978	88	228	0.507	1.571	0.637	0.808
9	0.436	192.695	27.77	118	234	0.457	1.744	0.574	0.737
10	0.208	181.884	30.092	99	228	0.509	1.876	0.533	0.845
	0.3488								

Average grain size: 0.3488 um^2

Recrystallization Zone Normalized/ Heat affected zone:

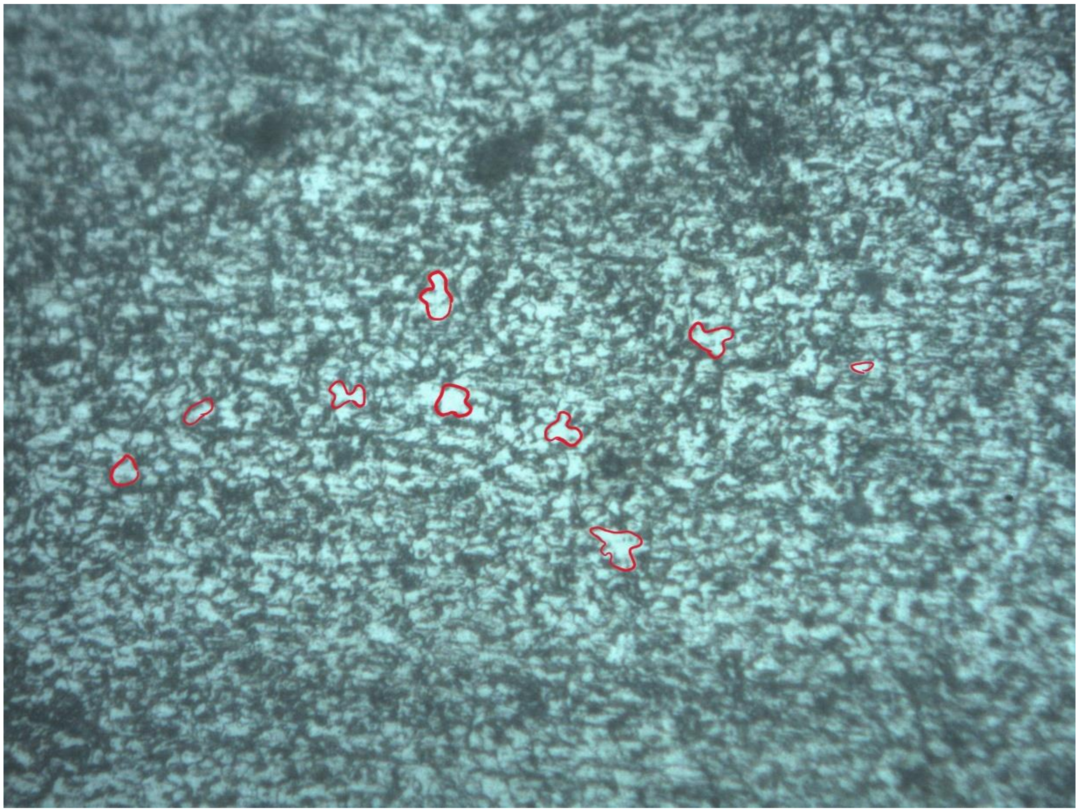


Figure 8:Area of 12 grains in R.Z.N

	Area(um^2)	Mean	StdDev	Min	Max	Circ.	AR	Round	Solidity
1	0.309	220.01	24.265	123	246	0.601	1.263	0.792	0.888
2	0.312	205.548	28.762	123	243	0.357	1.309	0.764	0.725
3	0.623	198.627	26.571	98	240	0.348	1.44	0.695	0.804
4	0.292	203.952	29.501	116	241	0.404	1.805	0.554	0.74
5	0.28	202.177	30.648	105	243	0.512	1.598	0.626	0.811
6	0.298	197.152	31.706	105	238	0.655	1.457	0.686	0.925
7	0.241	166.794	27.134	86	213	0.561	1.351	0.74	0.811
8	0.427	207.724	24.584	129	241	0.398	2.602	0.384	0.731
9	0.111	201.066	28.401	123	238	0.631	1.505	0.664	0.944
10	0.671	202.555	28.221	105	241	0.485	1.809	0.553	0.792
11	0.409	195.742	26.824	114	233	0.364	1.372	0.729	0.714
12	0.199	171.972	25.634	93	213	0.525	1.709	0.585	0.824
	0.347667								

Average grain size: 0.347667 um^2

Coarse Zone:

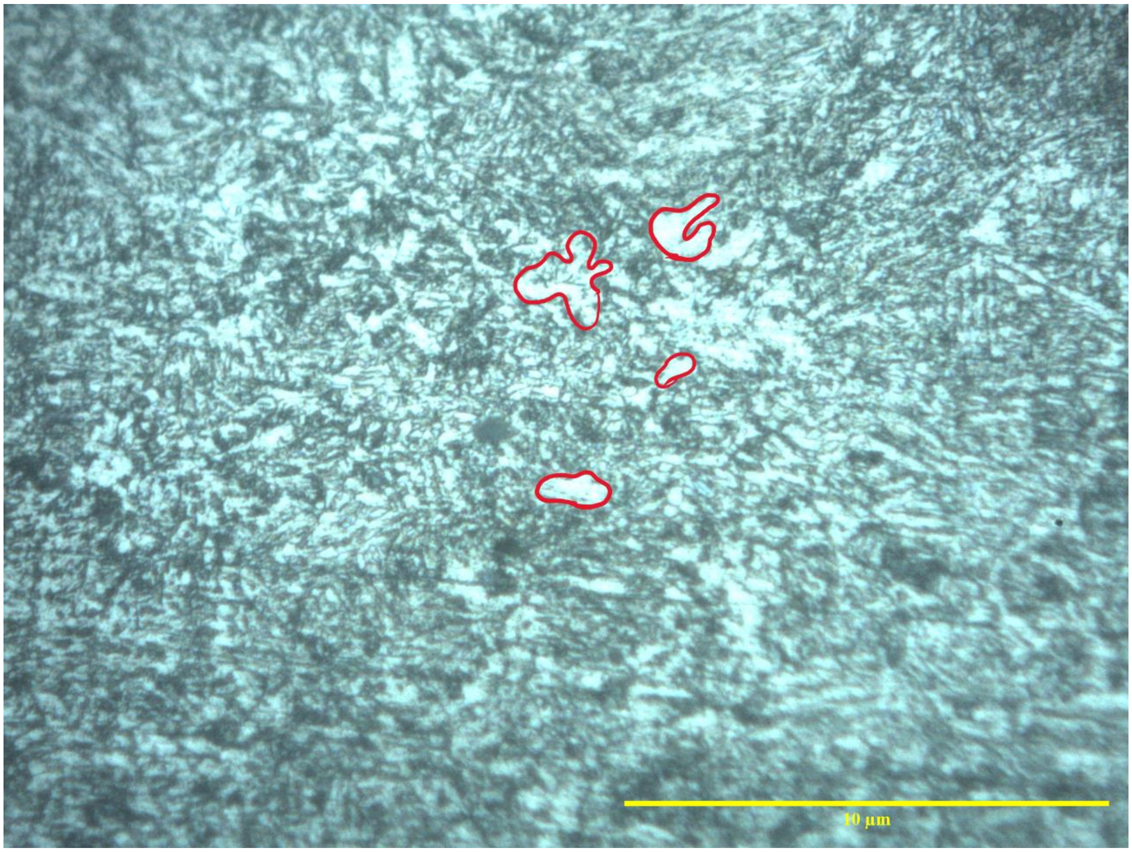


Figure 9: Area of 12 grains in coarse zone

	Area	Mean	StdDev	Min	Max	Circ.	AR	Round	Solidity
1	0.593	209.521	31.642	127	255	0.281	1.902	0.526	0.624
2	0.906	220.314	28.219	128	255	0.446	2.135	0.468	0.812
3	0.542	202.59	29.911	119	248	0.305	1.965	0.509	0.732
4	0.36	220.295	26.283	146	254	0.34	1.412	0.708	0.681
5	0.471	209.157	26.37	135	248	0.197	2.658	0.376	0.67
6	0.284	218.875	29.745	132	254	0.383	1.59	0.629	0.776
7	0.3	223.409	26.607	143	255	0.438	1.361	0.735	0.781
8	0.298	215.572	23.748	154	246	0.383	1.627	0.615	0.768
9	0.526	211.087	26.734	120	250	0.241	1.244	0.804	0.697
10	0.587	215.56	23.615	145	248	0.347	1.251	0.799	0.771
11	0.438	219.96	26.73	134	255	0.253	2.202	0.454	0.633
12	2.252	225.308	25.554	131	255	0.327	1.677	0.596	0.71
	0.62975								

Average grain size: 0.62975 um^2

Weld pool:

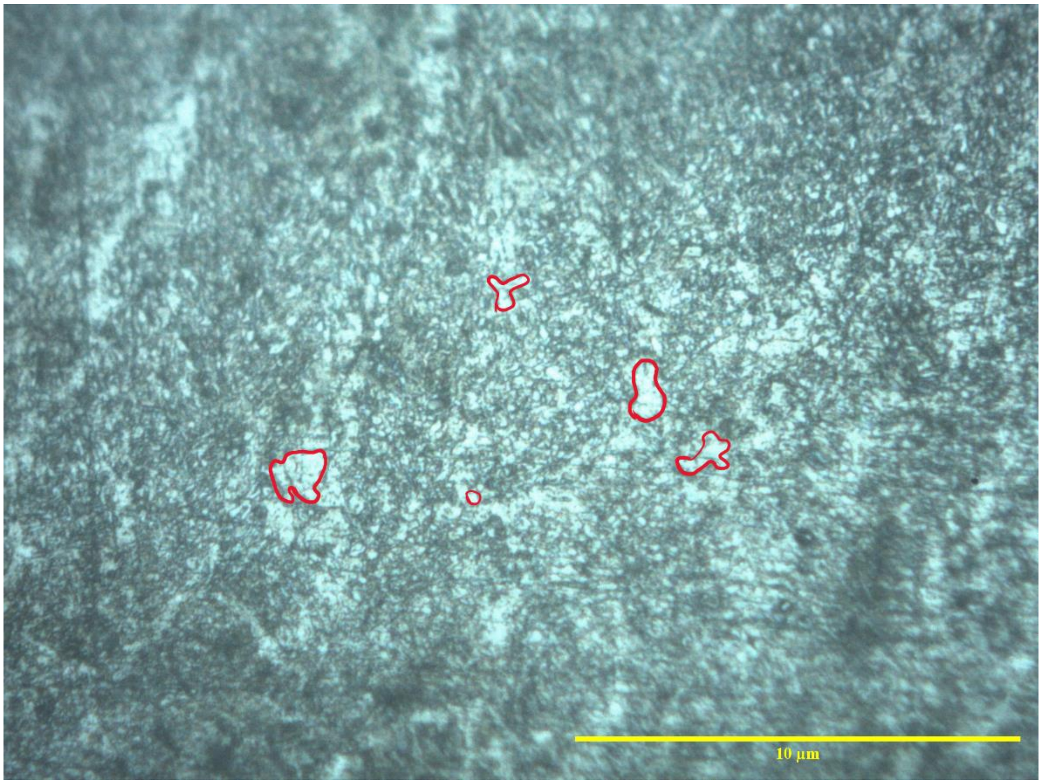


Figure 10: Area of 10 grains in weld pool

	Area	Mean	StdDev	Min	Max	Circ.	AR	Round	Solidity
1	1.604	212.02	29.509	111	255	0.28	2.202	0.454	0.701
2	0.602	202.542	30.843	120	252	0.307	1.388	0.72	0.75
3	0.404	194.387	33.738	109	253	0.273	1.632	0.613	0.687
4	0.79	205.44	28.458	116	251	0.263	1.099	0.91	0.745
5	0.789	203.323	28.238	125	251	0.239	1.492	0.67	0.594
6	0.427	205.459	30.623	122	252	0.325	1.904	0.525	0.764
7	1.12	207.98	30.18	108	254	0.321	1.087	0.92	0.803
8	0.183	208.023	27.507	128	245	0.525	2.334	0.428	0.785
9	0.261	207.433	26.78	114	243	0.351	1.986	0.504	0.723
10	0.69	187.135	28.496	110	247	0.327	1.764	0.567	0.712
	0.687								

Average of grain size= 0.687um^2

ASTM grain number:

We know ASTM Grain number indicates the number of grains one can see under 100x magnification of the microscope (in inches). So ASTM grain size number increases with decreasing grain size. The formula for ASTM Grain size is $N=2^{(G-1)}$ where N is the no of grains per square inch and G is the ASTM Grain Number. Solving this equation we get

$$G= (\log_{10}(N)/\log_{10}(2))+1$$

Zone	Area (um^2)	Area($inch^2$)	No of Grains	No of Grains per Square Inch (500×)	No of Grains per square inch (100x)	ASTM Grain Number, G= (log(N)/log(2))+1
Base metal	2.441	3.78×10^{-10}	8	2.1150033×10^{10}	5.28750825×10^{10}	36.62
Recrystallization Zone	4.478	6.95×10^{-9}	10	1438848921	3.597122302×10^{10}	36.066
Heat affected zone	4.172	6.47×10^{-9}	12	1855685503	4.639213758×10^{10}	36.433
Coarse Zone	7.557	1.171×10^{-9}	12	1019425406	2.548563515×10^{10}	35.569
Weld pool	6.87	1.0648×10^{-9}	10	939099403	2.347748509×10^{10}	35.45

The base metal has a higher ASTM grain number, while the weld pool and coarse zone have lower grain numbers. From the analysis, I learned that a higher grain number signifies smaller grain size, indicating finer grains. Therefore, the base metal has finer grains, whereas the coarse zone and weld zone have coarser grains. This is discussed in the discussion section.

Hardness Profile and Report:

Hardness profile with data table is presented below:

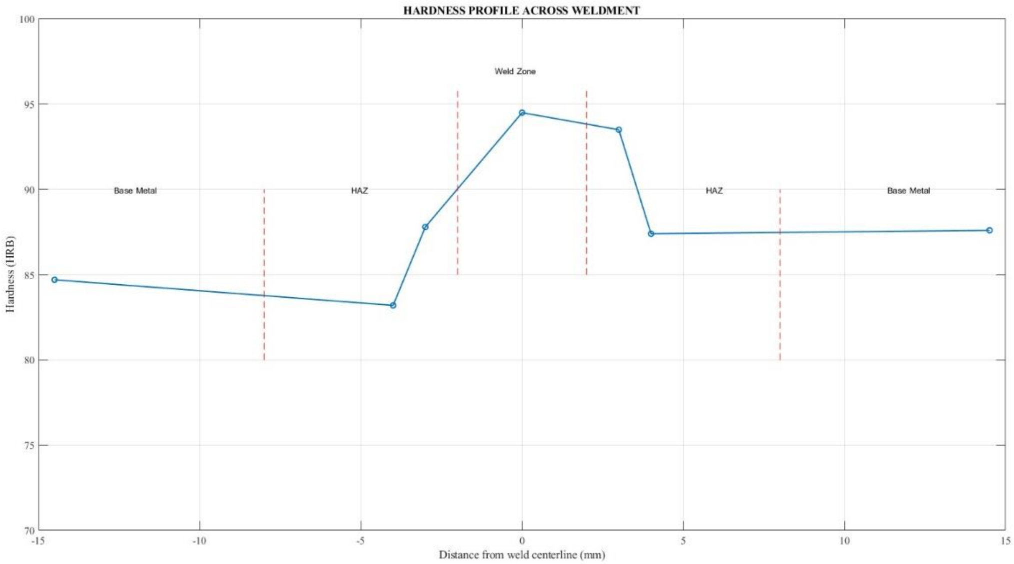


Figure 11: Hardness Profile

HRB HARDNESS TEST		
ZONE	**WELD CENTER (MM)**	**HARDNESS (HRB)**
BM	-14.5	84.7
NEAR HAZ	-4	83.2
HAZ	-3	87.8
WELD	0	94.5
HAZ	3	93.5
NEAR HAZ	4	87.4
BM	14.5	87.6

Hardness profile is related to the microstructure, which is discussed later. The MATLAB code for generating this profile is provided here. Additionally, In the discussion section I mentioned HAZ softening.

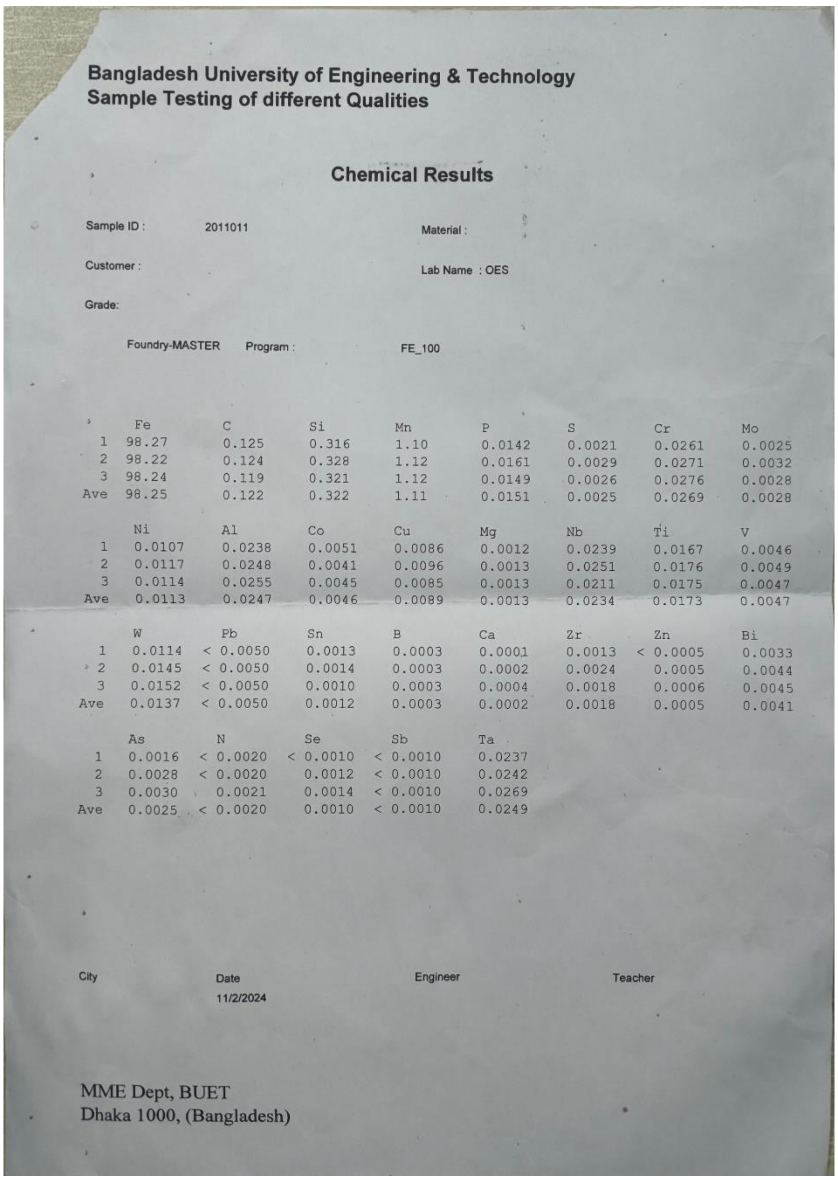
Hardness Report

Type of Test	**Hardness Test**
Sample Identification	Mild Steel (UNS K02709)
Type of Hardness Test	Rockwell B (100kg)
Parent Metal	Mild Steel
Thickness of material	5.5 mm
Type of Weld	Stringer bead

Welding Process	Gas Metal Arc Welding (GMAW) or MIG Welding
Consumables	Cu-coated mild steel Electrode
Heat Treatment or Ageing	Information not supplied
Comments	As per the client's requirement, the stringer bid was positioned in the middle and kept straight.
Witnessed by	Mr. Wasif Ur Rahman

OES Report and Carbon Equivalent Number:

The following data was received from the OES report:



Bangladesh University of Engineering & Technology
Sample Testing of different Qualities

Chemical Results

Sample ID : 2011011
Material :
Customer :
Lab Name : OES
Grade:
Foundry-MASTER Program : FE_100

	Fe	C	Si	Mn	P	S	Cr	Mo
1	98.27	0.125	0.316	1.10	0.0142	0.0021	0.0261	0.0025
2	98.22	0.124	0.328	1.12	0.0161	0.0029	0.0271	0.0032
3	98.24	0.119	0.321	1.12	0.0149	0.0026	0.0276	0.0028
Ave	98.25	0.122	0.322	1.11	0.0151	0.0025	0.0269	0.0028

	Ni	Al	Co	Cu	Mg	Nb	Ti	V
1	0.0107	0.0238	0.0051	0.0086	0.0012	0.0239	0.0167	0.0046
2	0.0117	0.0248	0.0041	0.0096	0.0013	0.0251	0.0176	0.0049
3	0.0114	0.0255	0.0045	0.0085	0.0013	0.0211	0.0175	0.0047
Ave	0.0113	0.0247	0.0046	0.0089	0.0013	0.0234	0.0173	0.0047

	W	Pb	Sn	B	Ca	Zr	Zn	Bi
1	0.0114	< 0.0050	0.0013	0.0003	0.0001	0.0013	< 0.0005	0.0033
2	0.0145	< 0.0050	0.0014	0.0003	0.0002	0.0024	0.0005	0.0044
3	0.0152	< 0.0050	0.0010	0.0003	0.0004	0.0018	0.0006	0.0045
Ave	0.0137	< 0.0050	0.0012	0.0003	0.0002	0.0018	0.0005	0.0041

	As	N	Se	Sb	Ta
1	0.0016	< 0.0020	< 0.0010	< 0.0010	0.0237
2	0.0028	< 0.0020	0.0012	< 0.0010	0.0242
3	0.0030	0.0021	0.0014	< 0.0010	0.0269
Ave	0.0025	< 0.0020	0.0010	< 0.0010	0.0249

City
Date 11/2/2024
Engineer
Teacher

MME Dept, BUET
Dhaka 1000, (Bangladesh)

From report, C=0.122; Mn=1.11; Si=0.322; Cr=0.0269; Mo= 0.0028; V=0.0047; Ni=0.0113; Cu=0.0089

We know, **CE = C + (Mn+Si)/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15**

So, CE= 0.3689

UNS Number and ASTM Standard:

From the values of different element quantities, my sample's specification is **ASTM A572 Grade 50 (UNS K02709)**. Its chemical composition typically ranges:

Carbon (C): ≤ 0.23% Manganese (Mn): ≤ 1.35% Silicon (Si): ≤ 0.40% Phosphorus (P): ≤ 0.04% Sulfur (S): ≤ 0.05% Copper (Cu): ≥ 0.20%

Which matches best to our sample. Besides hardness for in this type of steel grade is 85-90HRB. Our hardness value fluctuated around this range.

It is a high-strength low alloy steel (mild steel). This steel is used in structural applications such as bridges, buildings, and other load-bearing structures. It is known for its excellent strength-to-weight ratio and weldability.[1]

Discussion:

Here, I will describe the findings section by section:

Microstructure and Hardness:

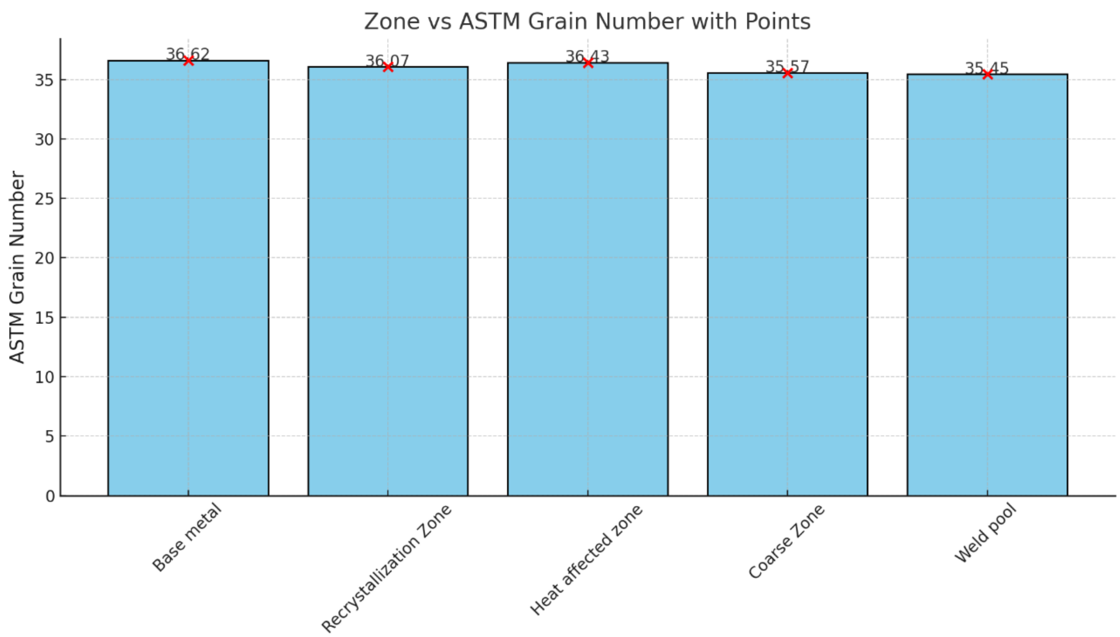


Figure 12:ASTM grain number vs Different Zones

A higher grain number indicates finer grains. Therefore, the lower grain number in the coarse zone indicates coarser grains. In my case, the weld zone created coarser grains than the coarse zone. Both the coarse zone and the weld zone have larger grains compared to the base metal, so the hardness should be lower. However, as observed in the hardness profile, the hardness increased in the coarse zone and reached its maximum in the weld zone. This is because the hardness value depends entirely on the indentation, which measures a small localized area. In my case, the area where the indentation was taken may have contained a martensitic structure, which is harder compared to ferrite and pearlite. Although this resulted in a higher hardness value locally, the overall hardness is not as high. [2]

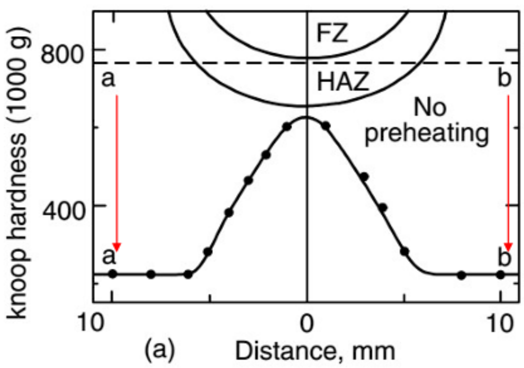


Figure 14:Hardness profile of 1040 steel (Due to martensitic structure)

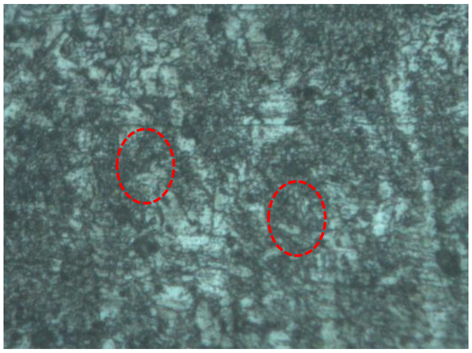


Figure 13: Martensitic Structure in weld zone

Different Cutting Process Selections:

First, I cut the sample from the very end of my workpiece using a hacksaw. A hacksaw was used to avoid the excessive heat generated by an electric saw, which could affect the microstructure. This piece was specifically prepared for microstructure analysis. For the 30mm OES sample, I used an electric saw to minimize my workload, as the excessive heat does not affect OES data. I performed hardness testing on the sample prepared for microstructure analysis.[3]

Reason for Choosing the Rockwell Hardness Testing Method:

The Rockwell hardness testing method is easier and faster compared to the Vickers hardness testing method. Although the Vickers method provides more accurate results, it requires proper surface preparation. Otherwise, it can result in significant errors, as small mistakes at the microscopic level can lead to large discrepancies. My sample was approximately 9mm, and in the transverse direction, it was not properly calibrated for the Vickers test. Therefore, I opted for the easier and quicker method, which was HRB.[4]

HAZ Softening:

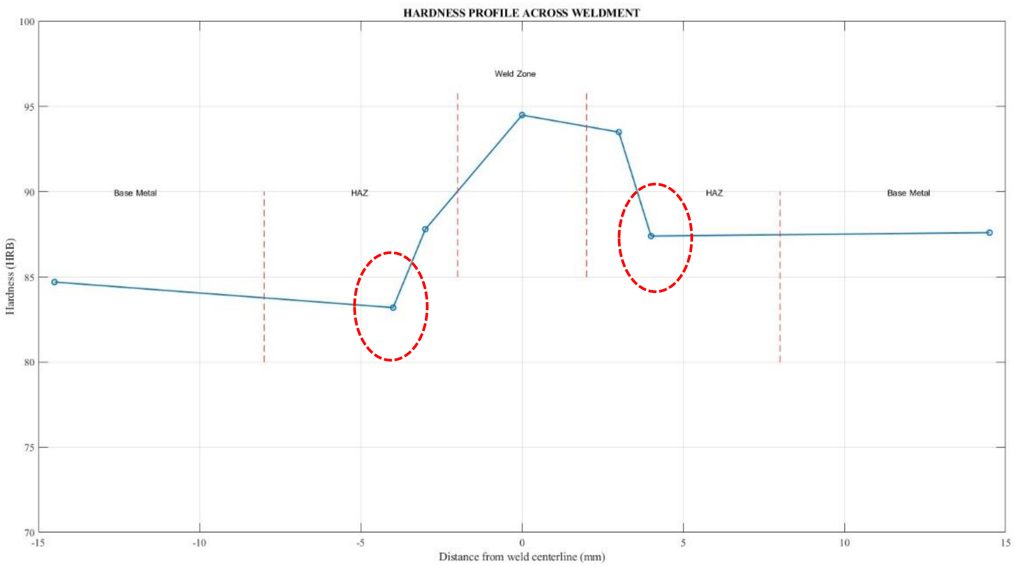


Figure 15: Hardness Profile indicating HAZ softening

In the heat-affected zone, the material is subjected to high temperatures below its melting point but sufficient to alter its microstructure. The closer the area is to the fusion boundary, the higher the peak temperature, which accelerates processes like recrystallization and grain growth. These changes reduce the hardness and strength of the material.

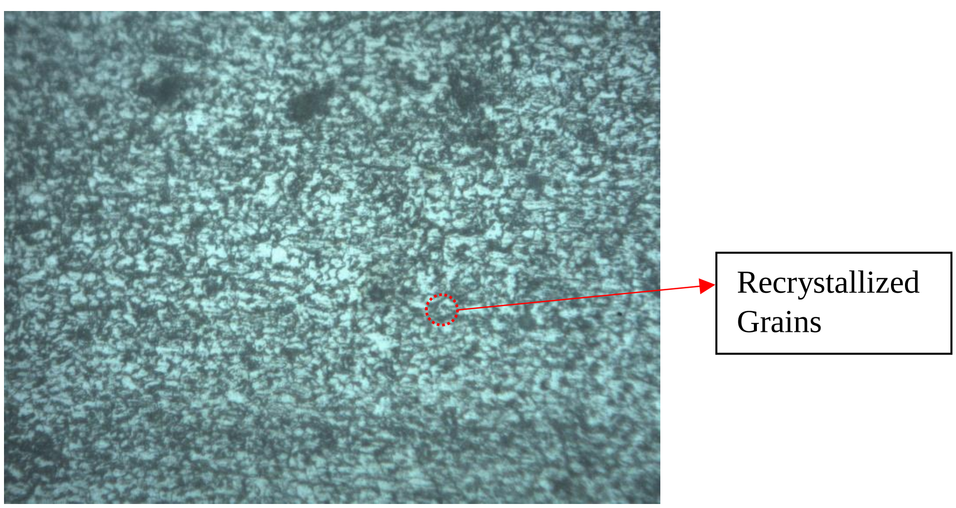


Figure 16:Microstructure of HAZ showing recrystallized grains

Hot Rolled/Cold Rolled?

My workpiece underwent hot rolling. The grain structure is small and fine, as observed in the base metal analysis. Additionally, all intermetallic inclusions and compounds, such as carbides, are aligned in one direction. Some recrystallized grains are clearly visible, indicating a typical hot-rolled microstructure.[5]

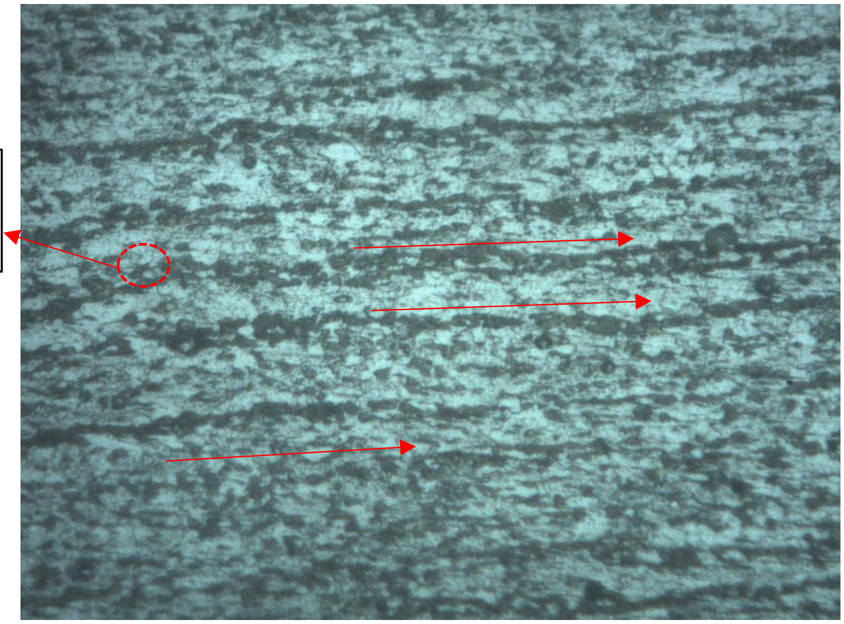


Figure 17:Base-metal indicating inclusion directions and recrystallized grains

Conclusion:

The experimental results demonstrate the significant impact of MIG welding on the mechanical and microstructural properties of mild steel. Key observations include variations in grain size, hardness profiles, and the formation of distinct zones, such as the heat-affected zone and the weld pool. The hardness values showed localized increases, reflecting microstructural changes like martensitic transformations. The findings confirm that precise control of welding parameters is essential to achieve desired material characteristics, ensuring reliability and performance in practical applications. This study underscores the importance of welding analysis in optimizing industrial processes and material design.

Welding Procedure Specification Form

Sample Welding Procedure Specification (WPS) Form

Welding Procedure Specification (WPS) No. 1101

Contractor: AKS Industry

Authorized by: Mr. Noah Jones — Revision No.: 2

Supporting PQR Identification: AWS D1.1 — Test Date: 10-11-24

WELDING PROCESS: ☐ FCAW-S ☐ FCAW-G ■ GMAW ☐ SMAW

WELD TYPE: ■ Groove ☐ Fillet

JOINT TYPE: ■ Direct Butt ☐ Indirect Butt ☐ T-Joint

Joint(s) Qualified (see Table 6.3):

Position: 1G — Groove Type:

Root Opening: — Root Face: — Groove Angle:

Backing: ☐ Yes ☐ No — Backing Type:

Backgouging: ☐ Yes ■ No — Backgouging Method:

TECHNIQUE: ■ Stringers ☐ Weave

ELECTRICAL CHARACTERISTICS

Current: ☐ AC ■ DCEP ☐ DCEN

Transfer Mode (GMAW): ☐ Short-circuiting ☐ Globular ■ Spray

BASE METAL

Material Specification: Mild Steel — Grade: ASTM A572

Welded to: Material Specification: Mild Steel — Grade: ASTM A572

Maximum Carbon Equivalent: 0.3689 — Bar size: 103.6L, 38.3W — Plate Thickness: 5.5mm

Coated Bar: ☐ Yes ■ No — Type of Coating:

FILLER METAL

AWS Specification: A5.17 — AWS Classification: EM12K

Describe filler metal (if not covered by AWS specifications): Cu coated mild steel

SHIELDING

Gas: ■ Single ☐ Mixture — Composition: Argon — Flow rate:

PREHEAT/INTERPASS

Preheat/Interpass Temperature (Min): — Interpass Temperature (Max):

WELDING PARAMETERS

Pass Number(s)	Electrode Diameter	Current: Type	Current: Amperage Range	Current: Volts Range	Current: Electrical Stickout	Travel Speed ipm [mm/min.]	Joint Detail
1	N/A	DCEP	100A	20V	N/A	N/A	Stringer Bead

Manufacturer or Contractor: AKS Industry

Authorized by: Mr. Noah Jones — Date: 10-11-24

Form A-2

PART-B MECHANICAL TEST

Transverse Tensile, Nick-Break, and Bend Test Report

A. Transverse Tensile Test

Sample Description	Specimen Number	Dimension	Tensile Strength, MPa	Rupture Location
Butt welded steel from pipe	1	Width: 2.46 cm Thickness: 0.86 cm Length: 30cm	404.117	At the weld center

B. Nick Break Test

Sample Description	Specimen Number	Dimension	Tensile Strength, MPa	Rupture Location
Butt welded steel from pipe	1	Width: 2.61cm Thickness: 0.87cm Width after nicking: 1.67cm	510.6541	At the weld center

C. Face Bend Test

Sample Description	Specimen Number	Dimension	Observation	Comment
Butt welded steel from pipe	1		No cracks, perfect root, no scratch	Good Sample

D. Impact Test

Sample Description	Specimen Number	Dimension	Energy	Comments
Butt welded steel from pipe	1	Thickness 0.62 cm Width 0.9 cm	15 J	Rupture at weld, Room temperature sample.

References:

[1]“Standard Specification for High-Strength Low-Alloy Columbium-Vanadium Structural Steel.” *A572/A572M*, www.astm.org/a0572_a0572m-21e01.html. Accessed 13 Dec. 2024.

[2]Rashed , D.M.A , (2024) , In: Metal Joining and Forming Processes .

[3]R. S. Chandel, "Effect of welding parameters and heat input on weld bead geometry," *International Journal of Advanced Manufacturing Technology*, vol. 16, no. 1, pp. 23-30, Jan. 2000. doi: 10.1007/BF02820865.

[4]G. E. Dieter, *Mechanical Metallurgy*, 3rd ed., McGraw-Hill, 1986, pp. 345-350.

[5]G. E. Dieter, *Mechanical Metallurgy*, 3rd ed. New York, NY, USA: McGraw-Hill, 1986, pp. 292-294.